MinimallySpeaking-SingleTimePoint

Stan West

2025-07-17

Data were collected from NDAR and classified as minimally speaking using SET CRITERIA. Here, we split the data according to age of CDI administration and are attempting to match 2-4, 4-6, and 6-8 year age bins. Networks based on these data may reveal different learning mechanisms in minimally speaking children as observed through network structure.

Data used here has first been filtered in several ways:

1. The oldest group (also the minimally speaking group) was filtered to include children from 72 months to 126 months. If a kid had more than one CDI administration, we took their first CDI (n = 51)
2. The middle and youngest groups only include children from 24 - 47 and 48 - 71 months, respectively and does not include any children also in the oldest group (n_youngest = 174,n_middle = 80).
3. The middle and youngest groups were then filtered to include appropriate entries given the standards set by the CDI (exclude WG forms where children were 19 months or older and produce more than 96 words & exclude WS forms where children were 30 months or older and produce more than 564 words) (n_youngest = 140 ,n_middle = 50).
4. The youngest group was filtered to not include any children from the middle group (n = 117).
5. Finally, the middle and youngest groups were filtered to only include the first CDI administration for any children with multiple CDIs (n_youngest = 108, n_middle = 38). The table below shows the distribution of multiple CDIs across groups.

group	n_entry	count
yng	1	113
yng	2	12
yng	3	1
mdl	1	26
mdl	2	12
old	1	31
old	2	11
old	3	8
old	4	1

Removed entries because of inappropriate forms

Some administrations were removed due to children exceeding the 50th percentile for their respective forms. These numbers are reported in the tables below.

Table 2: Words and Gestures Removed Subs

group	n
mdl	23
yng	27

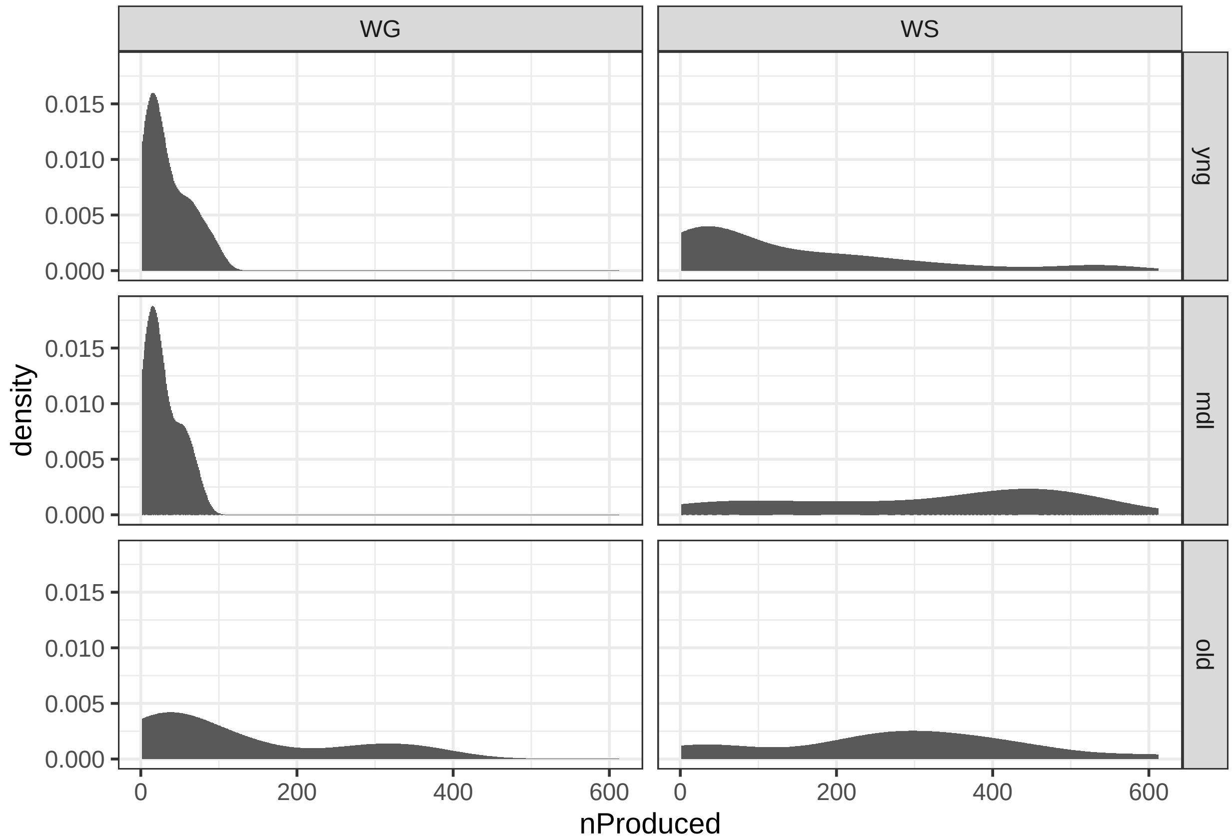
Table 3: Words and Sentences Removed Subs

group	n
mdl	33
yng	7

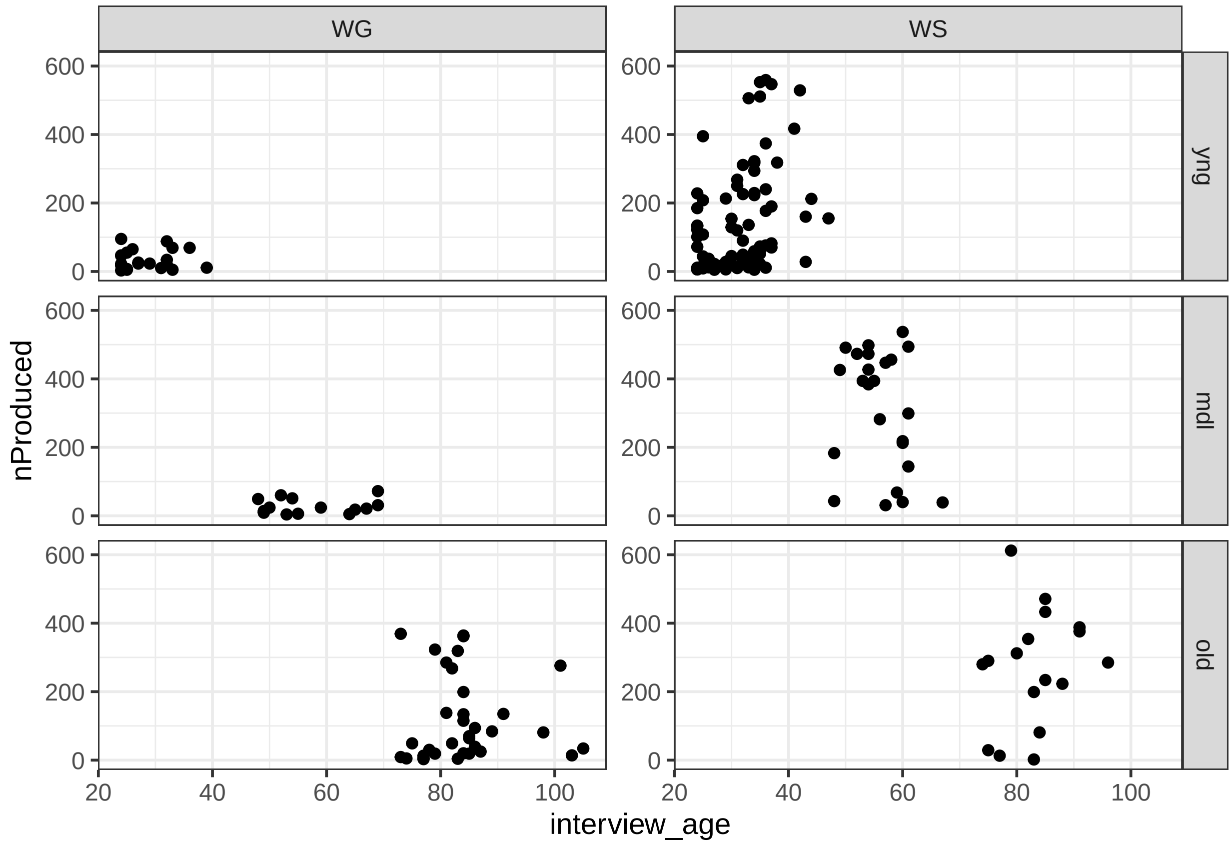
Tables of unmatched distributions across groups, vocabulary size, and age.

```
ggplot(all_subs, aes(x = nProduced))+
  geom_histogram(stat = "density")+
  facet_grid(group~form)+
  theme_bw()
```

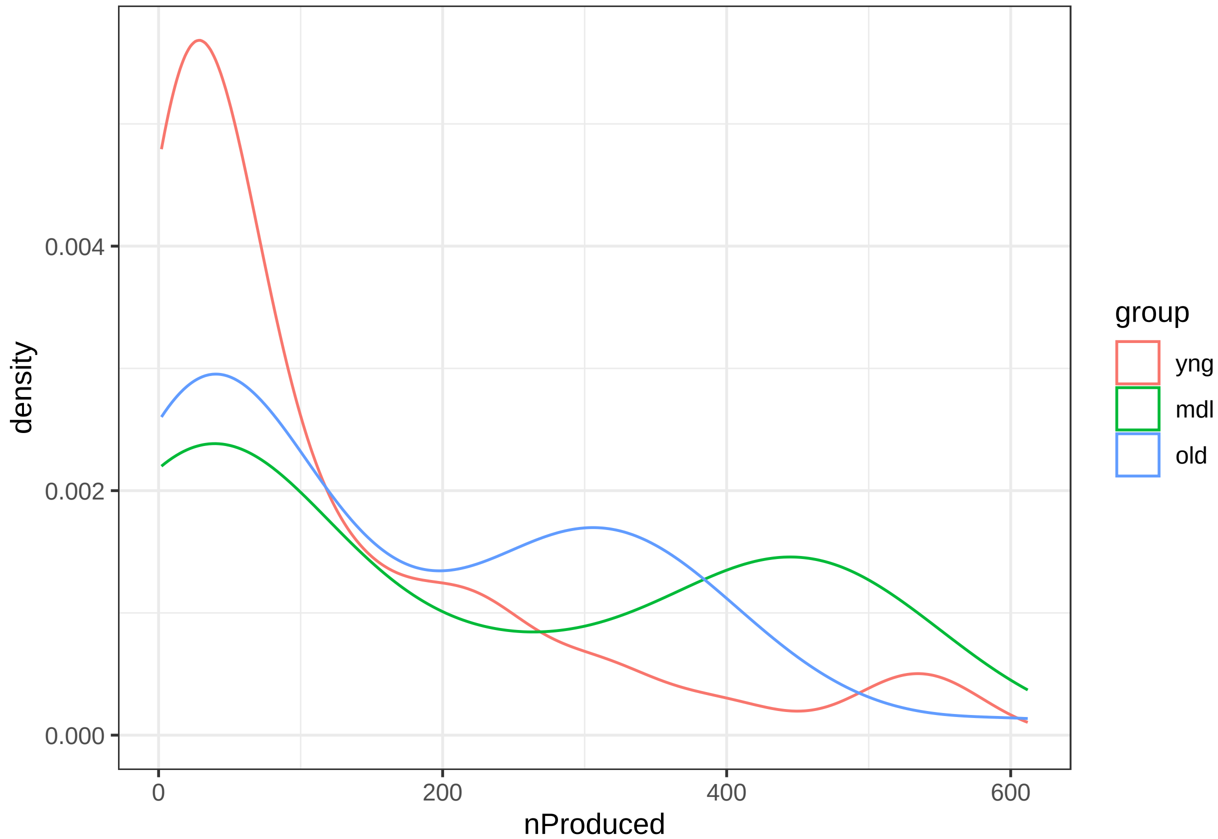
```
## Warning in geom_histogram(stat = "density"): Ignoring unknown parameters:
## ‘binwidth‘, ‘bins‘, and ‘pad‘
```




```r
ggplot(all_subs, aes(x = interview_age,y=nProduced))+
  geom_point()+
  facet_grid(group~form,axes = "all_y")+
  theme_bw()
```




```
ggplot(all_subs, aes(x =nProduced,color = group))+
  geom_density()+
  theme_bw()
```




```
all_subs |>
  group_by(group) |>
  summarize(n = n(),
            mean_production = mean(nProduced),
            mean_age = mean(interview_age)) |>
  kable(caption = "Descriptive Stats for Unmatched Sample")
```

Table 4: Descriptive Stats for Unmatched Sample

group	n	mean_production	mean_age
yng	108	115.6852	30.69444
mdl	38	206.3684	56.60526
old	51	168.6275	83.76471

Propensity matching

Either way you try matching, the fit is not great across the productive vocabulary distribution. Whether you match to the middle or old group, the difference in match is negligible.

Matching groups to the middle age group - For balanced groups, I matched to the smallest group first.

The procedure I am following here is first filtering down to the young and middle group and setting the middle group as the reference. Once those are matched, I fit another model matching the oldest group to the newly-selected young group that was just matched to the middle group. I combine the matched young-middle and young-old children and plot a density distribution across nProduced. Descriptives are also provided below.

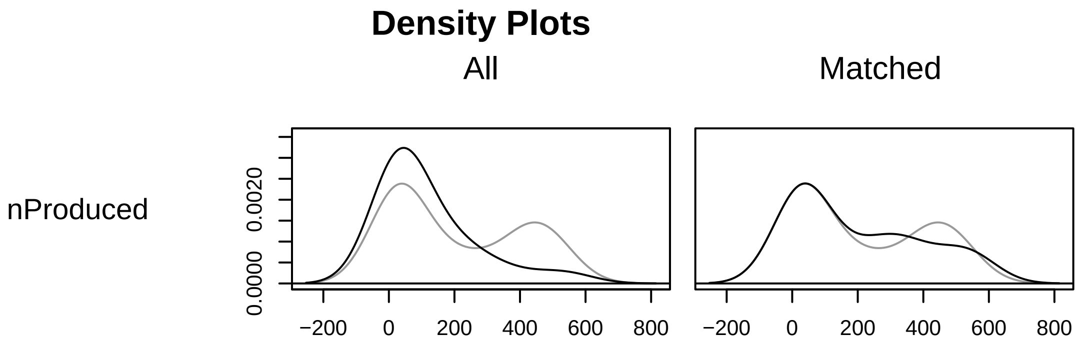
```
m.out_ym <- matchit(group ~ nProduced, data = all_subs |>
                   filter(group == 'yng'|group =="mdl") |>
                   mutate(group = factor(group,levels = c("mdl","yng"))),
                 method = "optimal",
                 distance = "glm")
```

```
## Warning: Fewer control units than treated units; not all treated units will get
## a match.
```

```
summary(m.out_ym)
```

```
##
## Call:
## matchit(formula = group ~ nProduced, data = mutate(filter(all_subs,
##     group == "yng" | group == "mdl"), group = factor(group, levels = c("mdl",
##     "yng"))), method = "optimal", distance = "glm")
##
## Summary of Balance for All Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance         0.7559        0.6938          0.6586      0.509     0.136
## nProduced      115.6852      206.3684         -0.6368      0.527     0.136
##           eCDF Max
## distance     0.268
## nProduced    0.268
##
## Summary of Balance for Matched Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance         0.6988        0.6938          0.0534     1.0014    0.0201
## nProduced      199.5526      206.3684         -0.0479     0.9881    0.0201
##           eCDF Max Std. Pair Dist.
## distance    0.1316          0.1637
## nProduced   0.1316          0.1414
##
## Sample Sizes:
##           Control Treated
## All            38     108
## Matched        38      38
## Unmatched       0      70
## Discarded       0       0
```

```
plot(m.out_ym, type = "density", interactive = TRUE,
     which.xs = ~nProduced )
```




```
df_ym <- match.data(m.out_ym)[c("subjectkey","interview_age","nProduced","form","group")] |>
  as.data.frame()


m.out_yo <- matchit(group ~ nProduced, data = rbind(df_ym, all_subs |>
                                                      filter(group == "old")) |>
                   filter(group == 'yng'|group =="old") |>
                   mutate(group = factor(group,levels = c('yng',"old"))),
                 method = "optimal",
                 distance = "glm")
```

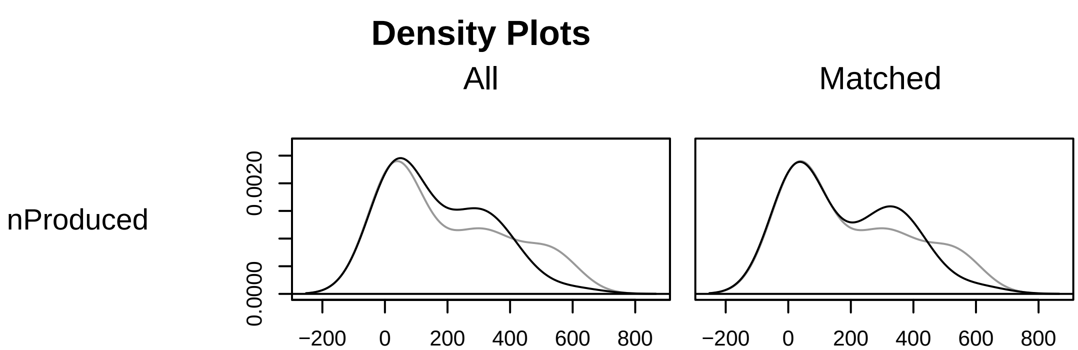
```
## Warning: Fewer control units than treated units; not all treated units will get
## a match.
```

```
summary(m.out_yo)
```

```
##
## Call:
## matchit(formula = group ~ nProduced, data = mutate(filter(rbind(df_ym,
##     filter(all_subs, group == "old")), group == "yng" | group ==
##     "old"), group = factor(group, levels = c("yng", "old"))),
##     method = "optimal", distance = "glm")
##
## Summary of Balance for All Data:
```

```
##             Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance          0.5764        0.5685          0.2008     0.6418    0.0542
## nProduced       168.6275      199.5526         -0.1973     0.6464    0.0542
##             eCDF Max
## distance      0.1517
## nProduced     0.1517
##
## Summary of Balance for Matched Data:
##             Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance          0.5734        0.5685          0.1247     0.7627    0.0319
## nProduced       180.3421      199.5526         -0.1225     0.7672    0.0319
##             eCDF Max Std. Pair Dist.
## distance      0.1316          0.1534
## nProduced     0.1316          0.1511
##
## Sample Sizes:
##             Control Treated
## All              38      51
## Matched          38      38
## Unmatched         0      13
## Discarded         0       0
```

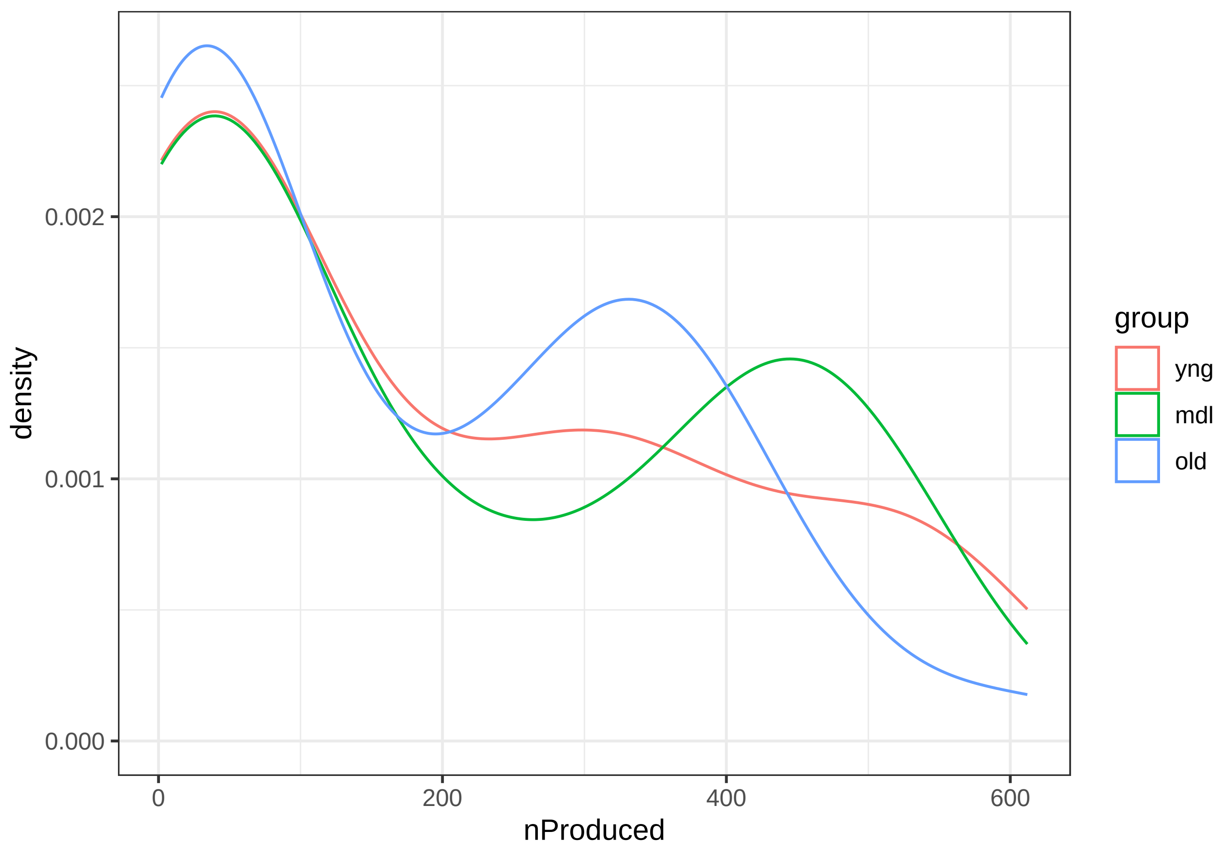
```
plot(m.out_yo, type = "density", interactive = TRUE,
     which.xs = ~nProduced )
```



```
df_o <- match.data(m.out_yo)[c("subjectkey","interview_age","nProduced","form","group")] |>
  filter(group == "old") |>
  as.data.frame()

df_matched_y <- rbind(df_ym, df_o) |>
  mutate(group = factor(group, levels = c("yng","mdl","old")))


ggplot(df_matched_y, aes(x = nProduced, color = group))+
  geom_density()+
  theme_bw()
```




```
df_matched_y |>
  group_by(group) |>
  summarize(n = n(),
            mean_age = mean(interview_age),
            sd_age = sd(interview_age),
            mean_produced = mean(nProduced),
            sd_produced = sd(nProduced)) |>
  as.data.frame() |>
  kable(col.names = c("group","n", "M(age)", "SD(age)", "M(produced)", "SD(produced)"))
```

group	n	M(age)	SD(age)	M(produced)	SD(produced)
yng	38	31.28947	5.093228	199.5526	194.9924
mdl	38	56.60526	6.056221	206.3684	196.1591
old	38	82.81579	6.366531	180.3421	170.7985

Matching groups to the old age group - Will result in over-representation of one of the groups

The procedure I am following here is first filtering down to the young and old group and setting the old group as the reference. Once those are matched, I fit another model matching the middle group to the newly-selected young group that was just matched to the old group. I combine the matched young-old and middle-young children and plot a density distribution across nProduced. Descriptives are also provided below.

```
m.out_yo <- matchit(group ~ nProduced, data = all_subs |>
                   filter(group == 'yng'|group =="old") |>
                   mutate(group = factor(group,levels = c("old","yng"))),
                 method = "optimal",
                 distance = "glm")
```

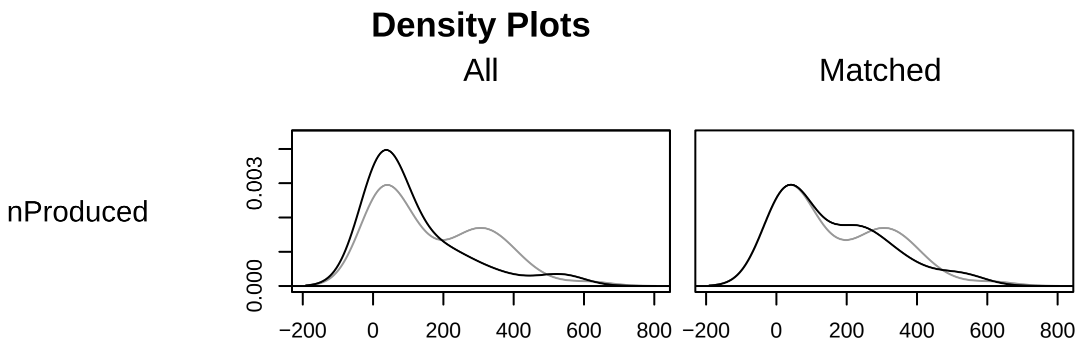
```
## Warning: Fewer control units than treated units; not all treated units will get
## a match.
```

```
summary(m.out_yo)
```

```
##
## Call:
## matchit(formula = group ~ nProduced, data = mutate(filter(all_subs,
##     group == "yng" | group == "old"), group = factor(group, levels = c("old",
##     "yng"))), method = "optimal", distance = "glm")
##
## Summary of Balance for All Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance         0.6880        0.6607          0.3636     0.8326    0.1152
## nProduced      115.6852      168.6275         -0.3718     0.8252    0.1152
##           eCDF Max
## distance    0.2151
## nProduced   0.2151
##
## Summary of Balance for Matched Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance         0.6651        0.6607          0.0585     0.9471    0.0201
## nProduced      160.5294      168.6275         -0.0569     0.9429    0.0201
##           eCDF Max Std. Pair Dist.
## distance    0.1176          0.1160
## nProduced   0.1176          0.1108
##
## Sample Sizes:
##           Control Treated
## All            51     108
## Matched        51      51
```

```
## Unmatched        0       57
## Discarded        0        0
```

```r
plot(m.out_yo, type = "density", interactive = TRUE,
     which.xs = ~nProduced )
```




```r
df_yo <- match.data(m.out_yo)[c("subjectkey","interview_age","nProduced","form","group")] |>
  as.data.frame()


m.out_my <- matchit(group ~ nProduced, data = rbind(df_yo, all_subs |>
                                                      filter(group == "mdl")) |>
                   filter(group == 'mdl'|group =="yng") |>
                   mutate(group = factor(group,levels = c('mdl',"yng"))),
                 method = "optimal",
                 distance = "glm")
```

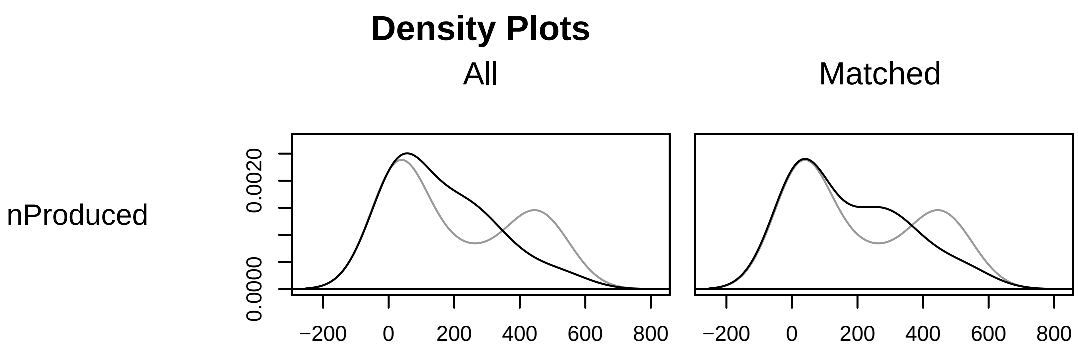
```
## Warning: Fewer control units than treated units; not all treated units will get
## a match.
```

```r
summary(m.out_my)
```

```
##
## Call:
```

```
## matchit(formula = group ~ nProduced, data = mutate(filter(rbind(df_yo,
##     filter(all_subs, group == "mdl")), group == "mdl" | group ==
##     "yng"), group = factor(group, levels = c("mdl", "yng"))),
##     method = "optimal", distance = "glm")
##
## Summary of Balance for All Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance         0.5806        0.5629          0.3072     0.5978    0.0732
## nProduced      160.5294      206.3684         -0.3011     0.6022    0.0732
##           eCDF Max
## distance    0.2441
## nProduced   0.2441
##
## Summary of Balance for Matched Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance         0.5752        0.5629           0.214     0.7310    0.0464
## nProduced      174.3947      206.3684          -0.210     0.7354    0.0464
##           eCDF Max Std. Pair Dist.
## distance    0.2105          0.2365
## nProduced   0.2105          0.2325
##
## Sample Sizes:
##           Control Treated
## All            38      51
## Matched        38      38
## Unmatched       0      13
## Discarded       0       0
```

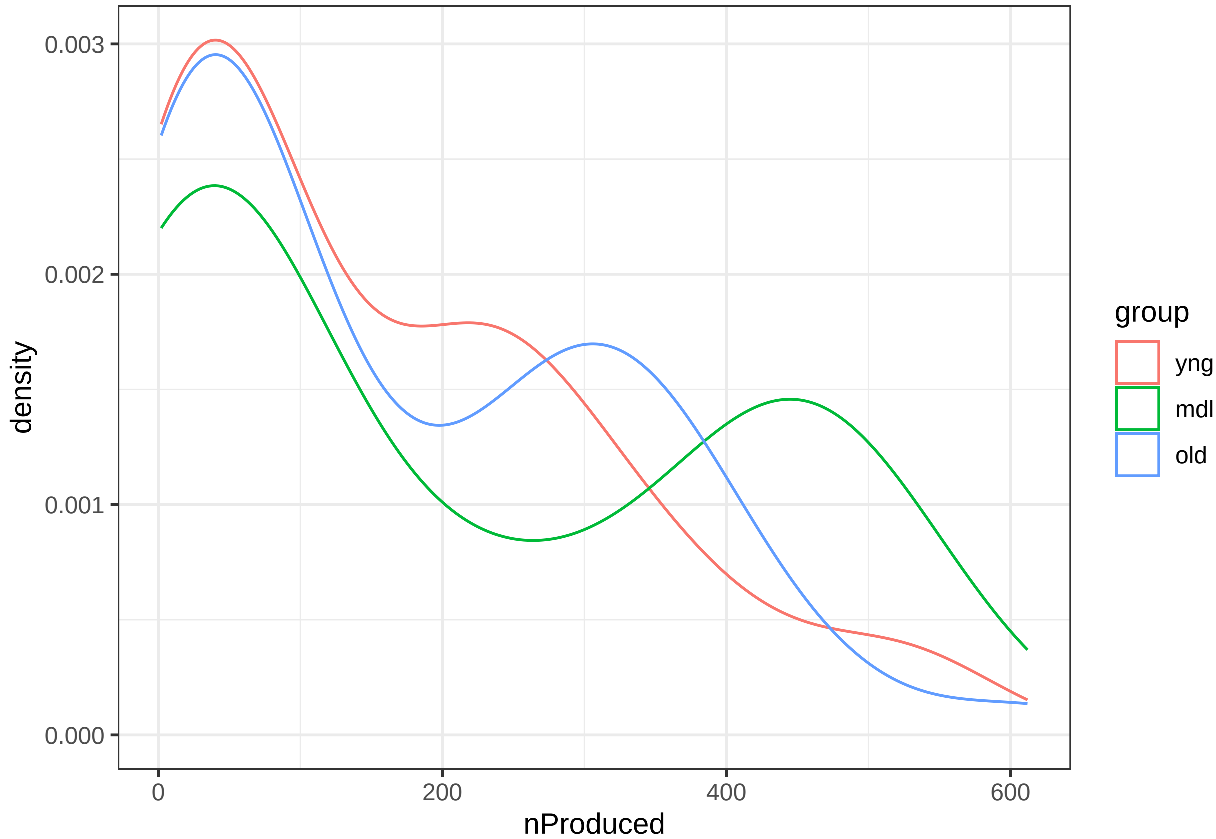
```r
plot(m.out_my, type = "density", interactive = TRUE,
     which.xs = ~nProduced )
```




```r
df_m <- match.data(m.out_my)[c("subjectkey","interview_age","nProduced","form","group")] |>
  filter(group == "mdl") |>
  as.data.frame()

df_matched_o <- rbind(df_yo, df_m) |>
  mutate(group = factor(group, levels = c("yng","mdl","old")))


ggplot(df_matched_o, aes(x = nProduced, color = group))+
  geom_density()+
  theme_bw()
```




```
df_matched_o |>
  group_by(group) |>
  summarize(n = n(),
            mean_age = mean(interview_age),
            sd_age = sd(interview_age),
            mean_produced = mean(nProduced),
            sd_produced = sd(nProduced)) |>
  as.data.frame()|>
  kable(col.names = c("group","n", "M(age)", "SD(age)", "M(produced)", "SD(produced)"))
```

group	n	M(age)	SD(age)	M(produced)	SD(produced)
yng	51	31.19608	5.272645	160.5294	152.2287
mdl	38	56.60526	6.056221	206.3684	196.1591
old	51	83.76471	7.322809	168.6275	156.7685

Propensity score matching with TriMatch - Mid Control

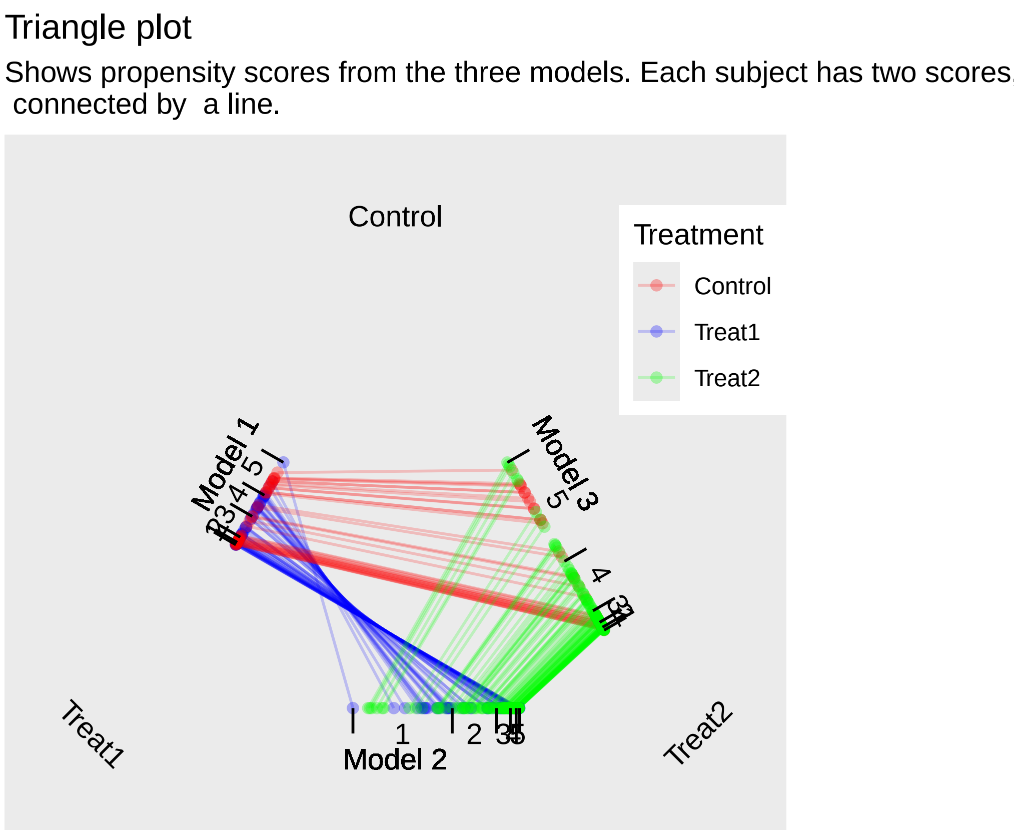
First, defining the formula to match only on nProduced. We then assign treatment and control labels to the data, here assigning the middle group as control and, oldest group as treatment 1, and youngest group as treatment 2. This logistic model will produce two propensity scores per subject. For the treatments, the scores will be acquired from a model matching to the control and the other treatment. The control will get two scores matching to each of the other two treatments. Three distance matrices are calculated corresponding to these propensity scores. A good match is determined by the size of a standardized distance metric for each pairwise propensity score. Using the caliper method, we can set different cutoffs for how many standard deviation distance units we want to include in our matching. The higher the caliper, the more liberal matching we are allowing.

This method has a hard time matching the higher end of productive vocabulary, so the more conservative matching estimates chop off the tail end of the nProduced distribution.

```
## Setting matching formula
formula <- ~ nProduced
all_subs$treats <- ifelse(all_subs$group == "mdl", "Control",
                          ifelse(all_subs$group == "old", "Treat1","Treat2"))
prop_scores <- trips(all_subs, all_subs$treats, formula)
```

```
## Using logistic regression to estimate propensity scores...
```

```
plot(prop_scores) + ggtitle("Triangle plot",subtitle = "Shows propensity scores from the three models.
```



Distribution and Descriptives with a 0.15 SD caliper cutoff.

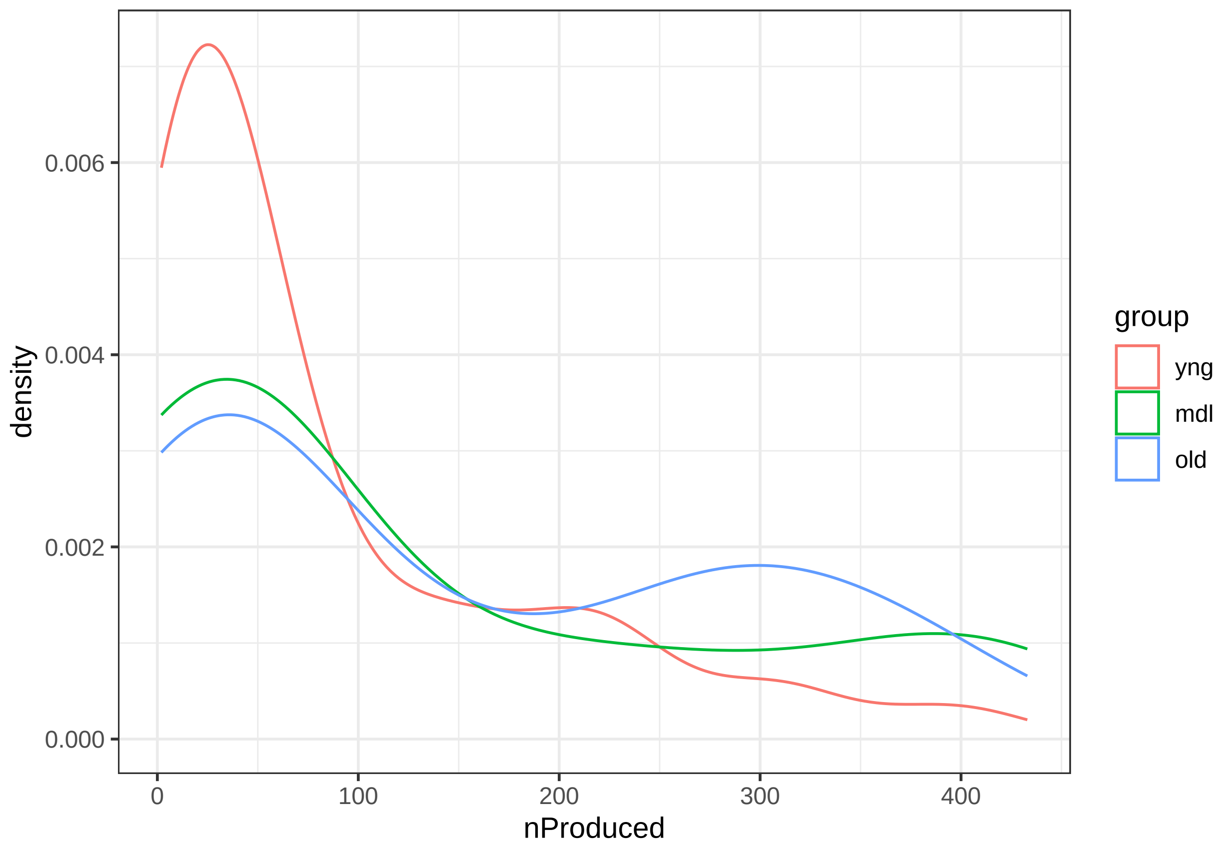
```
prop_scores <- prop_scores|>
  mutate(subjectkey = all_subs$subjectkey)

all_data.matched <- trimatch(prop_scores, method = NULL, caliper = 0.15) |>
  pivot_longer(cols = c("Treat1", "Control","Treat2"),
               names_to = "treat",
```

```
                values_to = "id",
                values_transform = as.integer) |>
  left_join(select(prop_scores,subjectkey,id), by = "id")

trimatched_subs <- all_subs |>
  filter(subjectkey %in% all_data.matched$subjectkey)


ggplot(trimatched_subs, aes(x = nProduced, color = group))+
  geom_density()+
  theme_bw()
```




```
trimatched_subs |>
  group_by(group) |>
  summarize(n = n(),
            mean_age = mean(interview_age),
            sd_age = sd(interview_age),
            mean_produced = mean(nProduced),
            sd_produced = sd(nProduced)) |>
  as.data.frame()|>
  kable(col.names = c("group","n", "M(age)", "SD(age)", "M(produced)", "SD(produced)"))
```

group	n	M(age)	SD(age)	M(produced)	SD(produced)
yng	97	30.50515	5.385404	86.73196	100.7302
mdl	30	56.83333	6.549985	132.43333	148.4624
old	47	83.87234	7.594669	149.95745	138.8765

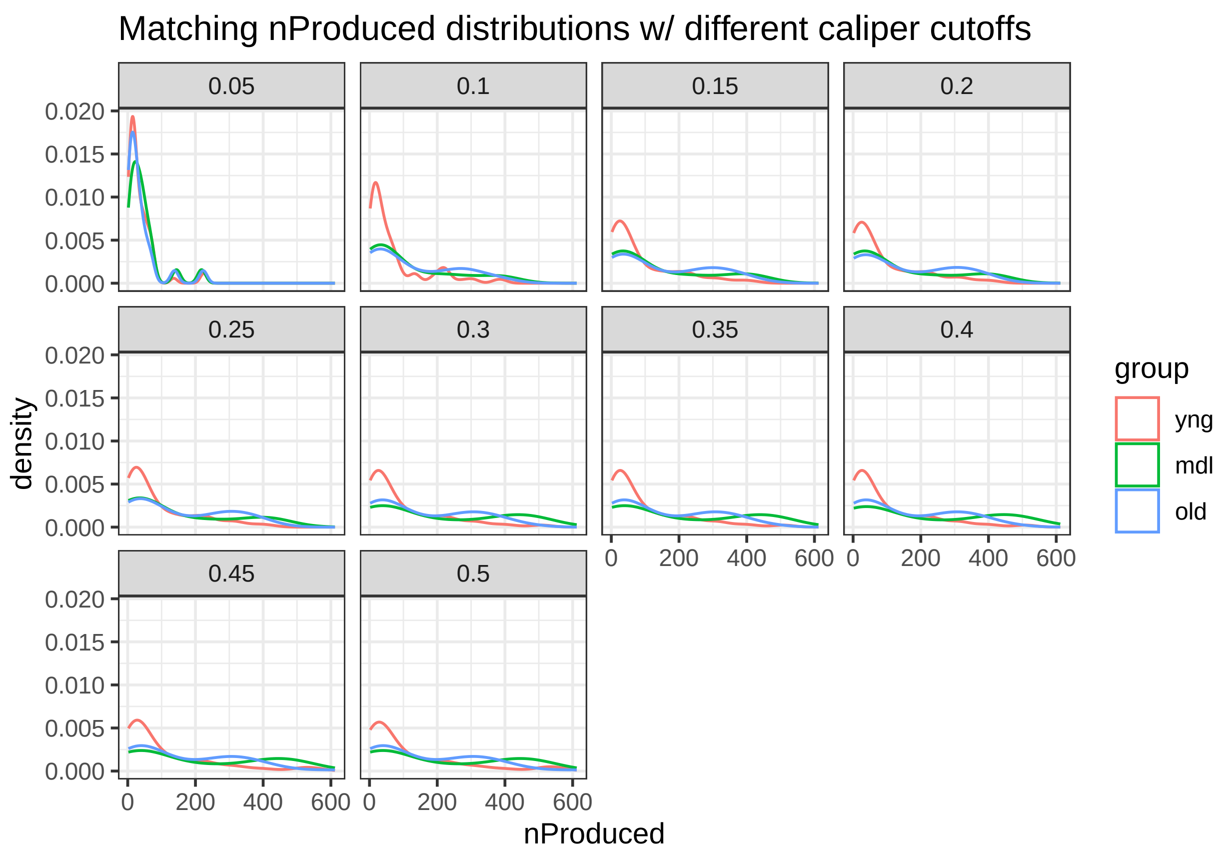
Series of calipers

Setting lower calipers (in standardized units) makes the matching more strict. More liberal calipers keeps larger distance propensity triplets. As you approach 0.5, you retain more and more data until you are left with the full unmatched sample at 0.5. Differences in distributions across caliper cutoffs seem to dissappear at around 0.25.

```
series <- map_dfr(.x = seq(0.05,0.5,0.05),function(x){
  d_matched <- trimatch(prop_scores, method = NULL, caliper = x) |>
    pivot_longer(cols = c("Treat1", "Control","Treat2"),
                 names_to = "treat",
                 values_to = "id",
                 values_transform = as.integer) |>
  left_join(select(prop_scores,subjectkey,id), by = "id")

  trimatched_subs <- all_subs |>
    filter(subjectkey %in% d_matched$subjectkey) |>
    mutate(caliper = x)
  return(trimatched_subs)
})

ggplot(series , aes(x = nProduced, color = group))+
  geom_density()+
  theme_bw()+
  facet_wrap(~caliper) +
  labs(title = "Matching nProduced distributions w/ different caliper cutoffs")
```




```
series |>
  group_by(caliper,group) |>
  summarize(n = n(),
            mean_age = mean(interview_age),
            sd_age = sd(interview_age),
            mean_produced = mean(nProduced),
            sd_produced = sd(nProduced)) |>
  as.data.frame()|>
  kable(col.names = c("caliper","group","n", "M(age)", "SD(age)", "M(produced)", "SD(produced)"))
```

```
## `summarise()` has grouped output by 'caliper'. You can override using the
## `.groups` argument.
```

caliper	group	n	M(age)	SD(age)	M(produced)	SD(produced)
0.05	yng	69	29.63768	4.636717	35.79710	40.98227
0.05	mdl	21	57.85714	7.143428	46.23810	50.46574
0.05	old	23	82.56522	8.095038	38.13043	50.45277
0.10	yng	87	29.87356	4.869947	71.64368	89.58145
0.10	mdl	28	57.21429	6.584815	111.42857	129.65579
0.10	old	41	83.92683	8.097500	125.68293	125.54769
0.15	yng	97	30.50515	5.385404	86.73196	100.73020
0.15	mdl	30	56.83333	6.549985	132.43333	148.46239
0.15	old	47	83.87234	7.594669	149.95745	138.87654

caliper	group	n	M(age)	SD(age)	M(produced)	SD(produced)
0.20	yng	99	30.48485	5.370675	89.32323	102.47995
0.20	mdl	30	56.83333	6.549985	132.43333	148.46239
0.20	old	49	83.83673	7.439610	153.40816	139.15368
0.25	yng	101	30.42574	5.356018	91.02970	102.70681
0.25	mdl	31	56.83871	6.439963	142.58065	156.51960
0.25	old	49	83.83673	7.439610	153.40816	139.15368
0.30	yng	104	30.43269	5.340384	99.09615	116.44796
0.30	mdl	37	56.51351	6.112926	197.43243	190.86274
0.30	old	50	83.86000	7.365141	159.76000	144.86497
0.35	yng	104	30.43269	5.340384	99.09615	116.44796
0.35	mdl	37	56.51351	6.112926	197.43243	190.86274
0.35	old	50	83.86000	7.365141	159.76000	144.86497
0.40	yng	104	30.43269	5.340384	99.09615	116.44796
0.40	mdl	38	56.60526	6.056221	206.36842	196.15912
0.40	old	50	83.86000	7.365141	159.76000	144.86497
0.45	yng	107	30.63551	5.424251	111.65421	136.74712
0.45	mdl	38	56.60526	6.056221	206.36842	196.15912
0.45	old	51	83.76471	7.322809	168.62745	156.76849
0.50	yng	108	30.69444	5.433469	115.68519	142.40745
0.50	mdl	38	56.60526	6.056221	206.36842	196.15912
0.50	old	51	83.76471	7.322809	168.62745	156.76849